GEBZE TECHNICAL UNIVERSITY

Computer Engineering Department

CSE 396 Module Documentation

Çılgın Kurye

Module M8: LED Animation and Physical Board

Company: Board&Byte **Group:** Group-8 **Date:** 6 November 2026

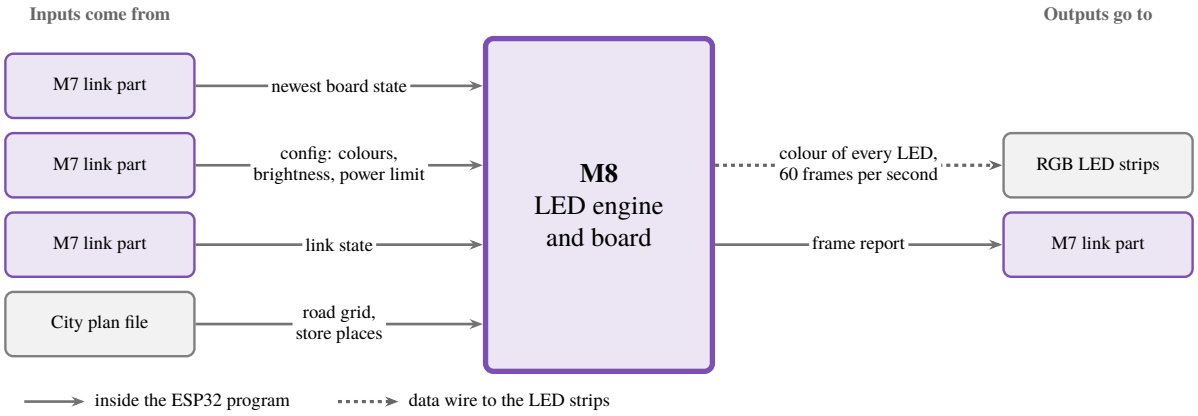
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1. Purpose

M8 builds the physical city board and writes the animation engine that turns the game state into light. The board is 60 × 60 cm. Programmable addressable RGB LED strips lie in all roads and in the four stores, and there are 3D printed buildings and places for the motion sensor and the vibration motors. In the game the board is the only map, so it must be clear to read from every side of the table.

Runs on	Technology	Requirements
ESP32 in the LED board, in one Arduino program together with M7, and the board itself for the hardware	C++ animation engine on the ESP32, programmable addressable RGB LED strips, an external power supply, 3D printing, a plywood base	R2 R5 R6 R7 R8 R11

M8 gets all its data from M7 in the same program and gives light to the LED strips. The figure shows the data on both sides.



2. Requirements covered

M8 takes part in six requirements. The last column says what will work when M8 is finished.

ID	Requirement	What M8 does for it	When M8 is done
R2	Delivery cycle	Delivery points blink in the player's colour, a flash when the package is delivered	Each player finds the own delivery point on the board
R5	No map on the phone	The board is the map with all motorcycles. "Find me" makes the player's dot white for 1 s	A player can always find the own motorcycle
R6	Obstacles	A red pattern for each type on the whole road segment, light blue for a puddle	Every obstacle type is clear from a distance
R7	Earthquake	A shock wave from the earthquake centre	The earthquake is seen on the table
R8	Physical LED board	60 × 60 cm, 6 roads in each direction, LEDs on all roads, printed buildings, four stores	The finished board stands on the table and shows the game
R11	Engineering requirements	Demo mode without a server, tested directly on the board with the ESP32	M8 can be built and tested without the server

3. Inputs and outputs

The board state comes 10 to 30 times per second, but M8 draws 60 frames per second, so it fills the time between two states by itself.

Direction	Data	From / to	Content
In	Board state	M7	Phase, players, delivery points, obstacles, earthquake
In	Config	M7	Player colours, brightness from 0.1 to 1, power limit
In	Link state	M7	Connected or searching
In	City plan	Shared file	Road grid and store places, turned into the LED map when the code is built
Out	Light	LED strips	The colour of every LED, 60 frames per second
Out	Frame report	M7	Frame rate and power estimate, shown to the admin

4. Main components

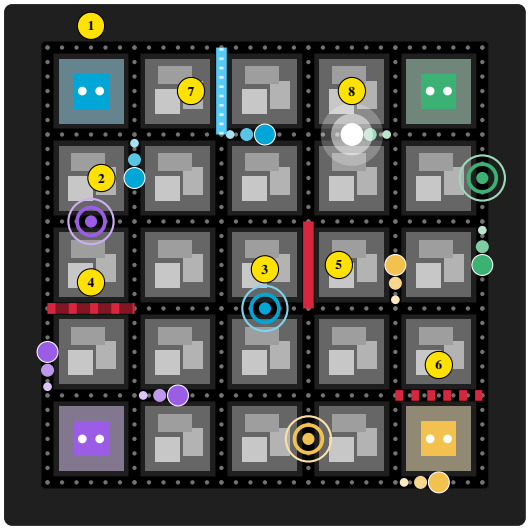
The engine has five parts. They run one after the other for every frame.

Component	Takes	Gives
LED map	City plan	The place of every LED on the roads and in the stores, made by a script
Motion smoother	The last two board states	Smooth motorcycle positions along the roads for every frame
Layers and effects	Board state, phase, link state	Five layers and the patterns for obstacles, earthquakes and game phases
Post processing and output	The mixed frame, config	Corrected colours inside the brightness and power limit, sent to the strips
Demo mode	A built in scenario	A full game look without a server

5. How it works

The board:

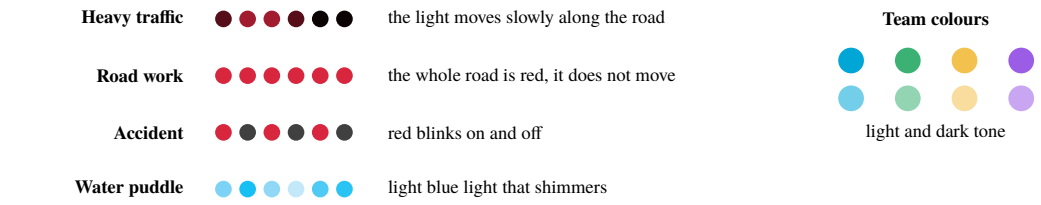
The base is plywood. On it there are 6 roads in each direction, so the city has 5 × 5 blocks. The LED strips lie in the roads between the crossings. There are no LEDs at the crossings, because two strips cannot lie on top of each other. The buildings are 3D printed from the same models as the 3D city on the phone, so a player finds the same building on the board and on the phone. The LEDs need an external power supply, and the software keeps the power under the limit in the config.



1 Team store in the team colour
2 Motorcycle: a bright dot with a short tail
3 Delivery point: blinks in the player's tone
4 Heavy traffic: a slow dim red wave
5 Road work: bright steady red
6 Accident: red blinking
7 Water puddle: light blue shimmer
8 "Find me": white flash for 1 second

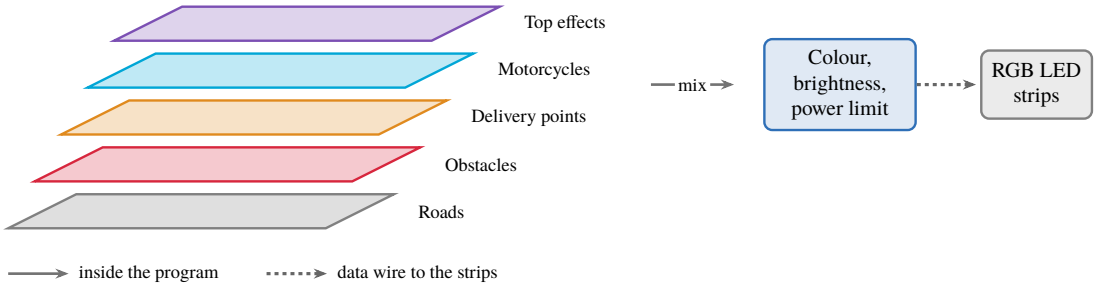
Colours:

There are four teams and each team has one colour: blue, green, yellow and purple. The two players of a team have a light and a dark tone of the team colour. Red is used only for obstacles, so red always means "this road slows you down". The pattern of the red tells the type of the obstacle. A water puddle is light blue, because it makes the road slippery, not slow. White is used only for "Find me", so a player always knows that the white dot is the own motorcycle.



Layers:

The engine draws 60 frames per second. Each frame is made from five layers, from the bottom to the top: the roads, the obstacles, the delivery points, the motorcycles and a top layer for short effects. A higher layer covers a lower one, so when two things fall on the same LED, the higher one is shown. When two motorcycles are on the same LED, their colours take turns. After the layers, the engine corrects the colours for the eye, sets the brightness from the config, and lowers all LEDs a little if the frame would need more power than the limit.

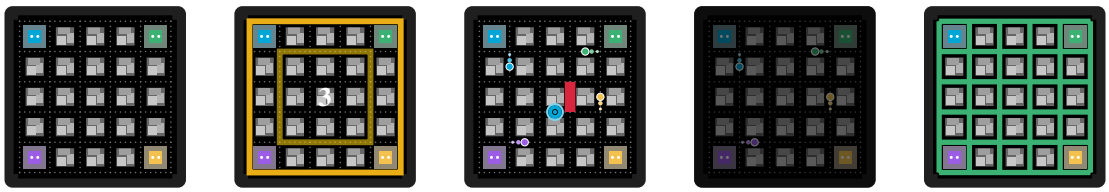


Smooth motion on the roads:

The board state comes less often than the frames, so the motion smoother moves each motorcycle dot between the last two states. It moves the dot along the roads, through the crossing, and never in a straight line over a block. The dot is drawn one state behind the newest one, so it adds a small delay. If a state is lost, the dot keeps moving along its road with its last speed until the next state comes. The dot has a short tail, so the direction of motion is easy to see.

Game phases and the link:

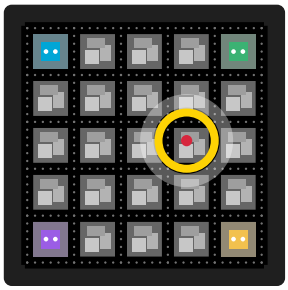
Every game phase has its own look, so the people around the table always know what happens.



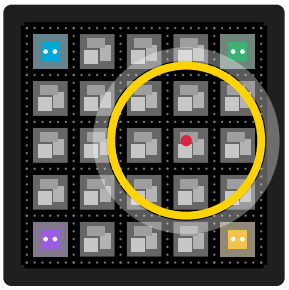
In the lobby the roads are dark and the four stores glow in the team colours. In the countdown the light moves from the edge to the centre in three steps, one for each number. During the game the board shows the live state. In a pause everything stays where it is, but all LEDs dim and breathe slowly. At the end all roads turn to the colour of the winner team, and in a draw the colours of the winners take turns. If M7 says that the link state is “searching”, the roads go dark and the stores breathe slowly in a dim white, so everybody can see that the board is waiting.

Earthquake:

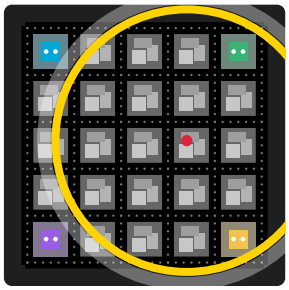
When an earthquake starts, a ring of light grows from the centre of the earthquake over the whole board, like a wave on water, and slowly fades. The road work after an earthquake is not a decision of M8. M1 puts it into the obstacle list, and M8 only draws it like any other obstacle until M1 removes it from the list.



start



a moment later



later

6. Testing

M8 is tested alone on the real board with the ESP32, without M7 and without the server. A built in demo scenario gives recorded board states as test data and plays a full game with every animation and every game phase. Test patterns check the order and place of the LEDs, the 8 player colours (three people must name each colour correctly), the power limit at full white and the heat in a long run.